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20A3105403

III B. Tech. I Semester Regular/Supplementary Examinations, Oct-2024

DESIGN AND ANALYSIS OF ALGORITHMS

(Common to CSE,IT and CSE-RL)

Time: 3 Hours**Dt: 30-10-2024****Max. Marks: 70**

Question paper contains **FIVE** units
Answer one question from each unit

Q.No.	Questions	Marks	BTL
UNIT I			
1. a)	Define the terms: i) Omega notation, ii) Theta notation	7M	1
b)	What is meant by performance analysis? Explain	7M	2
OR			
2. a)	What is an asymptotic notation? Explain different types of asymptotic notations with examples.	7M	3
b)	Discuss the rules to follow pseudo code with an example	7M	2
UNIT II			
3. a)	Write pseudo code for binary search with an example.	7M	2
b)	Explain general method of divide and conquer	7M	3
OR			
4. a)	Explain the average case time complexity of Quick sort	7M	2
b)	Explain finding minimum maximum using divide and conquer and pseudo code	7M	3
UNIT III			
5. a)	Explain minimum spanning tree using prim's algorithm with example and discuss time complexity with pseudo code	7M	4
b)	Write about single source shortest paths	7M	4
OR			
6. a)	Explain optimal merge patterns with an example and write psuedo code and discuss time complexity	7M	3
b)	Explain about minimum cost spanning trees with example	7M	3
UNIT IV			
7. a)	Construct an optimal travelling sales person tour using Dynamic Programming for the given data: $\begin{pmatrix} 3 & 2 & 9 & 3 \\ 5 & 0 & 6 & 2 \\ 9 & 6 & 0 & 7 \\ 7 & 2 & 5 & 0 \end{pmatrix}$	7M	2
b)	Describe the Dynamic 0/1 Knapsack problem. Find an optimal solution for the dynamic programming 0/1 knapsack instance for $n=3$, $m=6$, profits are $(p_1, p_2, p_3) = (1, 2, 5)$, weights are $(w_1, w_2, w_3) = (2, 3, 4)$.	7M	1

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OR			
8. a)	Find the all pairs shortest path solution for the graph represented by below adjacency matrix: $\begin{pmatrix} \infty & 6 & 5 & 4 \\ 3 & \infty & 2 & 6 \\ 18 & 6 & \infty & 7 \\ 8 & 12 & 10 & \infty \end{pmatrix}$	7M	3
b)	Explain Reliability Design problem with suitable example.	7M	2
UNIT V			
9. a)	What is a Hamiltonian Cycle? Explain how to find Hamiltonian path and cycle using backtracking algorithm.	7M	2
b)	Distinguish between backtracking and branch and bound techniques.	7M	2
OR			
10. a)	State N-Queens problem and solve 8-Queens problem using backtracking.	7M	2
b)	Explain the FIFO BB 0/1 Knapsack problem procedure with the knapsack instance for $n=4$, $m=15$, $(p_1, p_2, p_3, p_4) = (10, 10, 12, 18)$, $(w_1, w_2, w_3, w_4) = (2, 4, 6, 9)$. Draw the portion of the state space tree and find optimal solution.	7M	3

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III B; Tech. I Semester Regular/Supplementary Examinations, Oct-2024

DESIGN AND ANALYSIS OF ALGORITHMS

(CSE-RL)

Time: 3 Hours**Dt: 30-10-2024****Max. Marks: 70**

Question paper contains **FIVE** units
Answer one question from each unit

Q.No.	Questions	Mark s	BTL
UNIT I			
1. a)	Define the terms: i) Omega notation, ii) Theta notation నిబంధనలను నిర్వచించండి: i) ఒమేగా సంజ్ఞామానం, ii) థీటా సంజ్ఞామానం	7M	1
b)	What is meant by performance analysis? Explain పనితీరు విశ్లేషణ అంటే ఏమిటి? వివరించండి	7M	2
OR			
2. a)	What is an asymptotic notation? Explain different types of asymptotic notations with examples. అసింప్టోటిక్ సంజ్ఞామానం అంటే ఏమిటి? ఉదాహరణలతో వివిధ రకాల అసింప్టోటిక్ సంజ్ఞామానాలను వివరించండి.	7M	3
b)	Discuss the rules to follow pseudo code with an example ఒకతో సూడో కోడ్ ని అనుసరించడానికి నియమాలను చర్చించండి ఉదాహరణ	7M	2
UNIT II			
3. a)	Write pseudo code for binary search with an example. బైనరీ శోధన కోసం సూడో కోడ్ ను ఉదాహరణతో వ్రాయండి	7M	2
b)	Explain general method of divide and conquer విభజించి జయించే సాధారణ పద్ధతిని వివరించండి	7M	3
OR			
4. a)	Explain the average case time complexity of Quick sort త్వరిత క్రమబద్ధీకరణ యొక్క సగటు కేసు సమయ సంక్లిష్టతను వివరించండి	7M	2
b)	Explain finding minimum maximum using divide and conquer and pseudo code డివైడ్ అండ్ కాంకర్ మరియు సూడో కోడ్ ని ఉపయోగించి కనిష్ట గరిష్టాలను కనుగొనడాన్ని వివరించండి	7M	3
UNIT III			
5. a)	Explain minimum spanning tree using prim's algorithm with example and discuss time complexity with pseudo code ఉదాహరణతో ప్రిమ్ యొక్క అల్గోరిథం ఉపయోగించి కనిస విస్తీర్ణం గల చెట్టును వివరించండి మరియు సూడో కోడ్ తో సమయ సంక్లిష్టతను చర్చించండి	7M	4
b)	Write about single source shortest paths ఒక మూలం చిన్నదైన మార్గాల గురించి వ్రాయండి	7M	4
OR			
6. a)	Explain optimal merge patterns with an example and write psuedo code and discuss time complexity ఒక ఉదాహరణతో సరైన విలీన నమూనాలను వివరించండి మరియు సూడో కోడ్ వ్రాయండి మరియు సమయ సంక్లిష్టత గురించి చర్చించండి	7M	3
b)	Explain about minimum cost spanning trees with example	7M	3

UNIT IV

7. a)	Construct an optimal travelling sales person tour using Dynamic Programming for the given data: ఇచ్చిన డేటా కోసం డైనమిక్ ప్రోగ్రామింగ్‌ని ఉపయోగించి సరైన ట్రావెలింగ్ సేల్స్ పర్సన్ టూర్‌ను రూపొందించండి: $\begin{pmatrix} 3 & 2 & 9 & 3 \\ 5 & 0 & 6 & 2 \\ 9 & 6 & 0 & 7 \\ 7 & 2 & 5 & 0 \end{pmatrix}$	7M	2
b)	Describe the Dynamic 0/1 Knapsack problem. Find an optimal solution for the dynamic programming 0/1 knapsack instance for $n=3$, $m=6$, profits are $(p_1, p_2, p_3) = (1, 2, 5)$, weights are $(w_1, w_2, w_3) = (2, 3, 4)$. డైనమిక్ 0/1 నాప్‌సాక్ సమస్యను వివరించండి. $n=3$, $m=6$ కోసం డైనమిక్ ప్రోగ్రామింగ్ 0/1 నాప్‌సాక్ ఉదాహరణ కోసం సరైన పరిష్కారాన్ని కనుగొనండి, లాభాలు $(p_1, p_2, p_3) = (1, 2, 5)$, బరువులు $(w_1, w_2, w_3) = (2, 3, 4)$.	7M	1

OR

8. a)	Find the all pairs shortest path solution for the graph represented by below adjacency matrix: గ్రాఫ్ కోసం అన్ని జతల చిన్నదైన మార్గం పరిష్కారాన్ని కనుగొనండి దిగువ ప్రక్కనే ఉన్న మాతృక ద్వారా సూచించబడుతుంది: $\begin{pmatrix} \infty & 6 & 5 & 4 \\ 3 & \infty & 2 & 6 \\ 18 & 6 & \infty & 7 \\ 8 & 12 & 10 & \infty \end{pmatrix}$	7M	3
b)	Explain Reliability Design problem with suitable example. తగిన ఉదాహరణతో విశ్వసనీయత రూపకల్పన సమస్యను వివరించండి.	7M	2

UNIT V

9. a)	What is a Hamiltonian Cycle? Explain how to find Hamiltonian path and cycle using backtracking algorithm. హామిల్టోనియన్ సైకిల్ అంటే ఏమిటి? బ్యాక్‌ట్రాకింగ్ అల్గారిథమ్‌ని ఉపయోగించి హామిల్టోనియన్ పాత్ మరియు సైకిల్‌ను ఎలా కనుగొనాలో వివరించండి.	7M	2
b)	Distinguish between backtracking and branch and bound techniques. బ్యాక్‌ట్రాకింగ్ మరియు బ్రాంచ్ మద్దత్ తేడాను గురించండి మరియు కట్టుబడి పద్ధతులు.	7M	2

OR

10. a)	State N-Queens problem and solve 8-Queens problem using backtracking. N-క్వీన్స్ సమస్యను పరిష్కరించండి మరియు బ్యాక్‌ట్రాకింగ్ ఉపయోగించి 8-క్వీన్స్ సమస్యను పరిష్కరించండి.	7M	2
b)	Explain the FIFO BB 0/1 Knapsack problem procedure with the knapsack instance for $n=4$, $m=15$, $(p_1, p_2, p_3, p_4) = (10, 10, 12, 18)$, $(w_1, w_2, w_3, w_4) = (2, 4, 6, 9)$. Draw the portion of the state space tree and find optimal solution. FIFO BB 0/1 నాప్‌సాక్ సమస్య విధానాన్ని వివరించండి $n=4$, $m=15$ కోసం నాప్‌సాక్ ఉదాహరణతో, $(p_1, p_2, p_3, p_4) = (10, 10, 12, 18)$, $(w_1, w_2, w_3, w_4) = (2, 4, 6, 9)$. రాష్ట్ర స్థలం యొక్క భాగాన్ని గీయండి చెట్టు మరియు సరైన పరిష్కారం కనుగొనండి.	7M	3

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NRIA20**20A3105403**

III B. Tech. I Semester Supplementary Examinations, May-2023

DESIGN AND ANALYSIS OF ALGORITHMS

(Common to CSE and IT)

Time: 3 Hours**Max. Marks: 70**

Question paper contains **FIVE** units
 Answer one question from each unit

Q.No.	Questions	Marks	BTL
UNIT I			
1. a)	What is an asymptotic notation? Differentiate among various types of asymptotic notations with an example each.	7M	4
b)	Explain about the following types of analysis (i) Best (ii) average (iii) worst (iv) Average	7M	2
OR			
2. a)	Explain the time complexity of Merge sort. Discuss best, and worst time complexities	7M	3
b)	Give the Rules for writing Pseudo code and also write its pros and cons	7M	2
UNIT II			
3. a)	Differentiate between Divide and Conquer and Greedy method	7M	4
b)	Solve the following array of elements by using quick sort 5 7 16 13 24 62 36 51 93	7M	3
OR			
4. a)	Design algorithms for simple union, find and makeset operations.	7M	4
b)	Why do we need binary search? Explain with an example.	7M	2
UNIT III			
5. a)	Differentiate between prims and krushkals algorithm with an example each.	7M	4
b)	With an example explain about single source shortest path problem.	7M	2
OR			
6. a)	Find the optimal solution for the fractional knapsack problem making use of greedy approach. Consider: n = 4 m = 6 kg (w1, w2, w3, w4) = (3, 2, 10, 2) (p1, p2, p3, p4) = (15, 20, 30, 14)	7M	3
b)	Discuss about 2-way merge patterns.	7M	3
UNIT IV			

7. a)	Explain about OBST with an example	7M	3
b)	Explain about matrix chain multiplication problem with an example	7M	2
OR			
8. a)	Write an algorithm for all pairs shortest path problem	7M	2
b)	Explain Reliability Design problem with suitable example.	7M	3
UNIT V			
9. a)	State N-Queens problem and solve 8-Queens problem using backtracking.	7M	3
b)	Explain about TSP using branch and bound with an example	7M	2
OR			
10. a)	Distinguish between backtracking and branch and bound techniques.	7M	4
b)	What is a Hamiltonian Cycle? Explain how to find Hamiltonian path and cycle using backtracking algorithm.	7M	3

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III B. Tech. I Semester Regular Examinations, Oct-2022

DESIGN AND ANALYSIS OF ALGORITHMS

(Common to CSE and IT)

Time: 3 Hours

Max. Marks: 70

Question paper contains **FIVE** units
Answer one question from each unit

Q.No.	Questions	Marks
UNIT I		
1. a)	Explain about Space Complexity and Time Complexity.	7M
b)	Discuss about Big Oh notation and Omega Notation.	7M
OR		
2. a)	Explain Pseudo code Conventions in detail.	7M
b)	Explain about Analysis Framework briefly.	7M
UNIT II		
3. a)	Contrast the Quick Sort with Merge Sort.	7M
b)	Explain Divide and Conquer in detail.	7M
OR		
4. a)	Explain Iterative and Recursive searching techniques in Binary Search.	7M
b)	Explain about Divide and Conquer applications in detail.	7M
UNIT III		
5. a)	Explain Knapsack Problem in Greedy Method.	7M
b)	What is Minimum Cost Spanning Trees? Discuss.	7M
OR		
6. a)	Explain about Kruskal's Algorithm briefly.	7M
b)	Discuss Optimal Merge Patterns with examples.	7M
UNIT IV		
7. a)	Explain about Travelling Sales Person Problem with an example.	7M
b)	What is the need of Reliability Design in Dynamic Programming? Discuss.	7M
OR		
8. a)	Discuss about All-Pairs Shortest Paths.	7M
b)	Explain 0/1 knapsack problem briefly.	7M
UNIT V		
9. a)	Explain the concept of Graph Coloring.	7M
b)	Explain about LC and FIFO Branch and Bound solutions.	7M
OR		
10.a)	Discuss about Hamiltonian Cycles in detail.	7M
b)	Contrast Back tracking Vs. Branch and Bound techniques.	7M

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NRIA20**20A3242401**

III B. Tech. II Semester Regular Examinations, March-2023

DESIGN AND ANALYSIS OF ALGORITHMS

(Common to CSE(AIML) and CSE(DS))

Time: 3 Hours**Max. Marks: 70**

-Question paper contains **FIVE** units
Answer one question from each unit

Q.No.	Questions	Marks	BTL
UNIT I			
1. a)	Define algorithm. Explain the characteristics of an algorithm.	7M	2
b)	Write an algorithm for finding the maximum element in an array of elements and show its time complexity.	7M	2
OR			
2. a)	Write recursive algorithm to find out nth Fibonacci number.	7M	3
b)	What is space complexity? With suitable example, explain how it is computed.	7M	2
UNIT II			
3. a)	Briefly explain Merge Sort Algorithm with suitable example and Derive its Time Complexity.	7M	2
b)	Design algorithms for simple union and find operations. Also discuss problems associated with these algorithms.	7M	3
OR			
4. a)	Sort the records with the following index values in ascending order using quick sort algorithm. 65,70,75,80, 33, 60,55, 22, 50,45, 11	7M	3
b)	Show how binary search algorithm works for searching 151, -14 and 9 in the following set of elements: -15, -6, 0, 7, 9, 23, 54, 82, 101, 112, 125, 131, 142, 151	7M	3
UNIT III			
5.	Explain the Job Sequencing with deadlines problem with appropriate example.	14M	2
OR			
6. a)	Describe single source shortest path algorithm with an example.	7M	2
b)	Differentiate between prim's and Kruskal's algorithms.	7M	2
UNIT IV			
7.	Construct the OBST with the identifier set(a1,a2,a3,a4) = (end, goto, print, stop) with (P1,P2,P3,P4) = (0.5,0.1,0.02,0.5) & (Q1,Q2,Q3,Q4,Q5) = (0.5,0.15,0.2,0.1,0.2) Also compute the cost of the tree.	14M	3
OR			
8. a)	Consider three stages of a system with r1=0.3, r2=0.5, r3=0.2 and c1=30, c2=20, c3=30 Where the total cost of the system is C=80 and u1=2, u2=3, u3=2 find the reliability design.	7M	3
b)	Write an algorithm for matrix chain multiplication problem using dynamic programming.	7M	2
UNIT V			
9. a)	Write an algorithm for graph colouring with an example using backtracking method.	7M	2
b)	Define the terms Branch and Bound. Explain about its general method.	7M	2
OR			
10. a)	Explain the principles of (a) Bounding. (b) FIFO Branch & Bound.	14M	2

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J. Gaur
3/3/2023